

K6 Load Testing: A Beginner’s Guide

Understanding performance testing with k6

AGENDA

- Why load testing, and why k6?
- Core K6 concepts
- Key metrics to watch
- Demo

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WHY K6 (AND NOT JMETER)?

- k6 treats a load test as code (JS/TS, versionable in Git) — JMeter treats it as a GUI-built XML config
- Easier to code-review, diff, and integrate into CI/CD pipelines
- Lightweight: k6's Go-based engine runs far more virtual users per machine than JMeter's Java/JVM threads
- Trade-off: JMeter still offers wider built-in protocol support (JDBC, JMS, LDAP, SOAP) and a GUI some QA teams prefer
- Bottom line: k6 fits teams that write code and want tests living next to the app

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WHY LOAD TESTING?

- Functional tests answer: “Does it work?”
- Load tests answer: “Does it work under pressure, for many users, fast enough?”
- Real-world failures often happen at scale, not on a single request
- Goal: find problems before your users do

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SMOKE VS LOAD VS STRESS: SIDE BY SIDE

	Smoke	Load	Stress
Question	Does it work at all?	Does it handle normal traffic?	Where does it break?
VUs	1	0→10→0 (steady)	0→20→50→100→0 (aggressive)
Duration	Very short	Medium, incl. steady state	Progressively heavier
Pass/Fail gate?	Yes (basic checks)	Yes (thresholds)	No — exploratory only
When to run	First, always	Before release / regularly	Occasionally, to know limits



CORE K6 CONCEPTS

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VIRTUAL USER (VU)

- A VU = one simulated user running your test script in a loop
- vus: 10 → 10 “people” hitting your app at the same time
- Analogy: 10 employees clicking through the website simultaneously, non-stop

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ITERATION

- One full run of the default function = 1 iteration
- If a VU runs 5 times, that’s 5 iterations
- More iterations = more real requests generated during the test

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CHECK

- A per-request validation: “Was this response OK?”
- Doesn’t stop the test if it fails — just records pass/fail
- Example: status is 200, response time < 500ms, body not empty
- Analogy: a checklist checkbox — one box unchecked, you keep going, but it’s on record

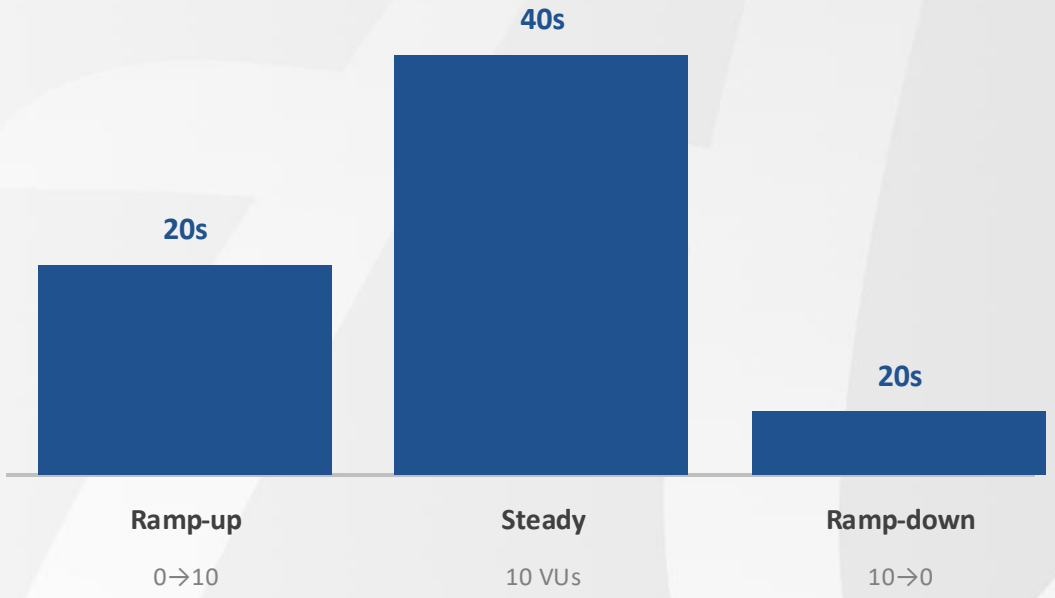
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THRESHOLD

- A pass/fail condition for the entire test
- If not met → whole test run is marked FAILED (non-zero exit code)
- Example: 95% of requests must finish under 800ms
- This is what blocks a CI/CD pipeline when performance regresses

STAGES (RAMP-UP / RAMP-DOWN)

- Instead of a fixed number of users, VUs increase and decrease over time
- Mimics real traffic: users don’t all arrive at once
- Pattern: ramp-up → steady state → ramp-down



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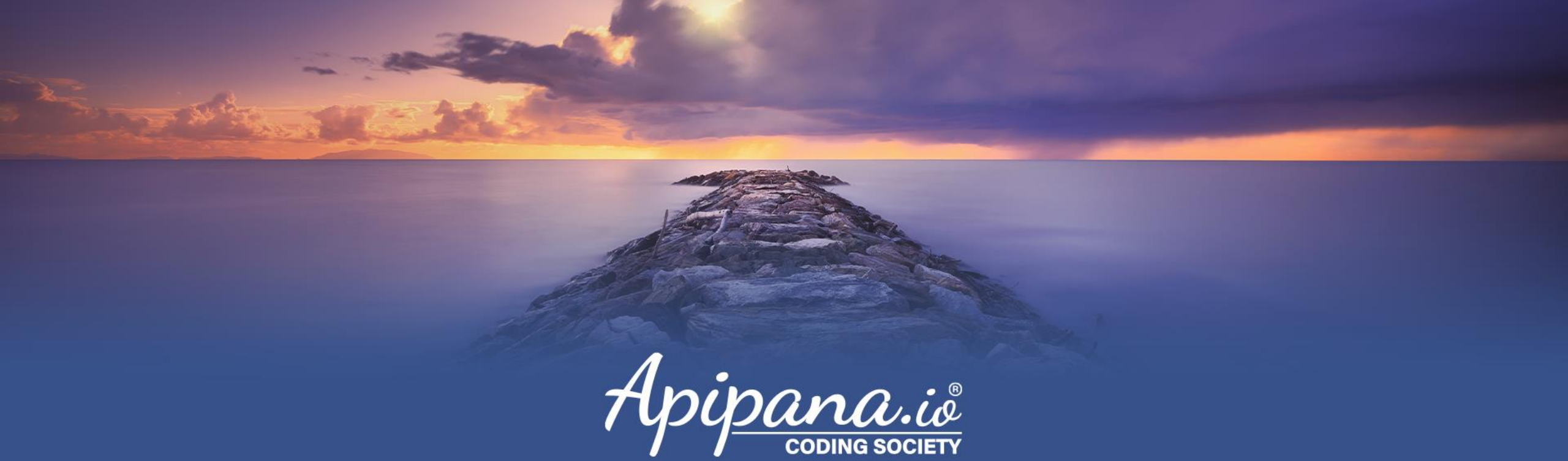
KEY METRICS TO WATCH

- http_req_duration — latency (look at p95, p99, not just average)
- http_req_failed — error rate
- vus — active virtual users over time
- iterations — total completed test runs

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CORE CONCEPTS CHEATSHEET

Concept	What it is	Example
VU	One simulated user running the script in a loop	vus: 10
Iteration	One full run of the default function	1 VU × 5 iterations = 5 requests
Check	Per-request validation (doesn't fail the test)	status is 200
Threshold	Pass/fail condition for the whole test	p(95)<800
Stages	VUs ramping up/down over time	0 → 10 → 0



Questions?

github.com/geeksusma/k6-mentoring